**7** A student connects the circuit shown in Figure 7.1.

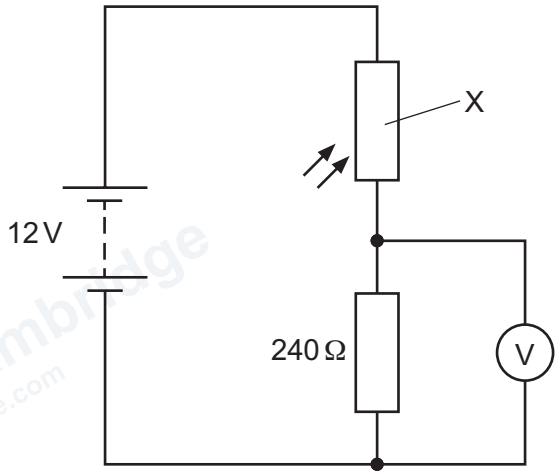


**Figure 7.1**

**(a)** The connecting wires in the circuit are made of metal.

State the name of the particles that flow in the metal wires.

..... [1]

**(b)** A bright light shines on component X.

**(i)** State the name of component X.

..... [1]

**(ii)** The resistance of component X is 400 Ω when the light shines on it.

Calculate the reading on the voltmeter.

reading on the voltmeter = ..... V [3]

**(c)** The brightness of the light shining on component X is reduced. The resistance of component X increases.

State the effect of this change on:

the current in the fixed resistor .....

the reading on the voltmeter. .....

[2]

[Total: 7]



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